

Qualify One Configured Record Source

Make stationary evidence attributable and terminal

E0 COPIED REPLAY — NO SENSOR, BUS, PIN, OR CALIBRATION CLAIM

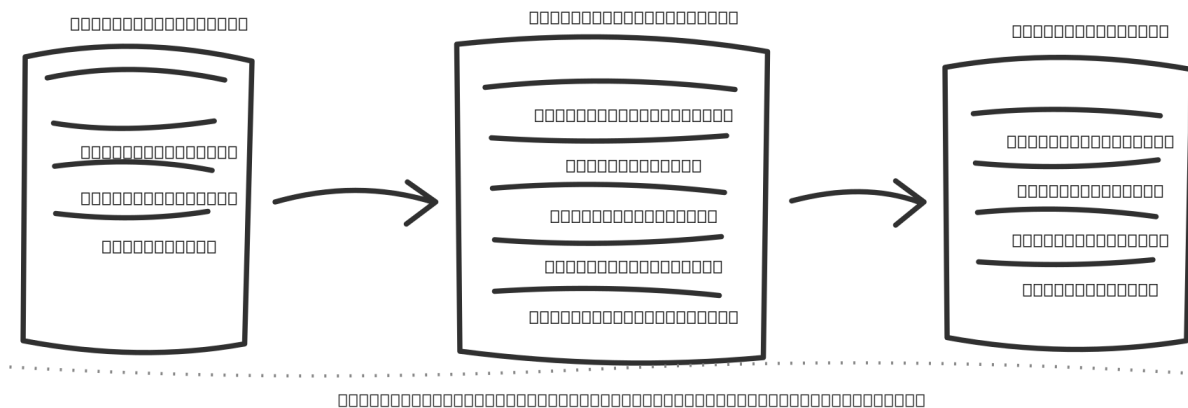
Lesson 068 — component
Prerequisite Lesson 067 records
Input one configured copied stream

Time 150–210 minutes
Board compile-only Mega replay
Observation named evidence cells

Responsibility. `InertialRecordQualificationPolicy` maps and evaluates a bounded stationary window from exactly one configured copied record source. It publishes one attributable terminal evidence envelope. It does not discover, probe, select, vote, fail over, configure, or physically calibrate a sensor.

1 Draw the one-source boundary

The configured domain includes source kind, model and ID; source configuration and calibration revisions; ranges; record schema and normalization revisions; qualification revision; mapping; sample count; age and gap limits; and stationary bounds. Every record must match the whole domain.

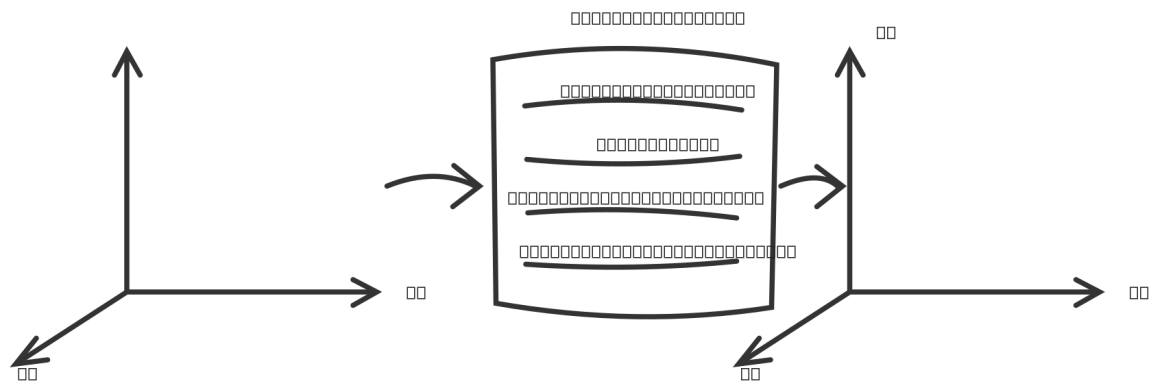


Alternative text: one complete copied source card enters one configured qualification desk and yields one copied terminal evidence card; the desk is explicitly separated from all physical hardware.

E0 establishes	E0 does not establish
exact-domain record comparison	physical device identity or support
proper-rotation mapping	actual mounting or native sensor axes
bounded stationary-trace checks	accuracy, bias, drift, or calibration
deterministic terminal evidence	I2C, voltage, pins, or data-ready timing
Prediction: if only the calibration revision changes, should the record enter the window? _____. Explain: _____.	

2 Map both vectors through one proper rotation

`SourceAxisMapping` chooses one signed source axis for each destination axis. Each source axis appears exactly once and the transform must be right-handed. Exactly 24 of all 216 signed-axis triples pass. The same mapping applies to acceleration and angular rate.



Alternative text: a hand-drawn source coordinate frame passes through a mapping card into a qualification coordinate frame; the card lists distinct axes, right handedness, shared vector mapping, and checked signed negation.

Duplicate axes and left-handed reflections are invalid configuration. Negating `INT32_MIN` is not representable and rejects the candidate without partial output or signed-overflow behavior. Lesson 044 continues to own pitch/roll presentation; Lesson 068 owns no angle.

Mapping	Result	Reason
+X, +Y, +Z	valid	identity proper rotation
+Y, -X, +Z	valid	right-handed quarter turn
+X, +X, +Z	invalid	source X used twice
+X, +Y, -Z	invalid	left-handed reflection

Map (10, -20, 30) through (+Y, -X, +Z): (_____, _____, _____).

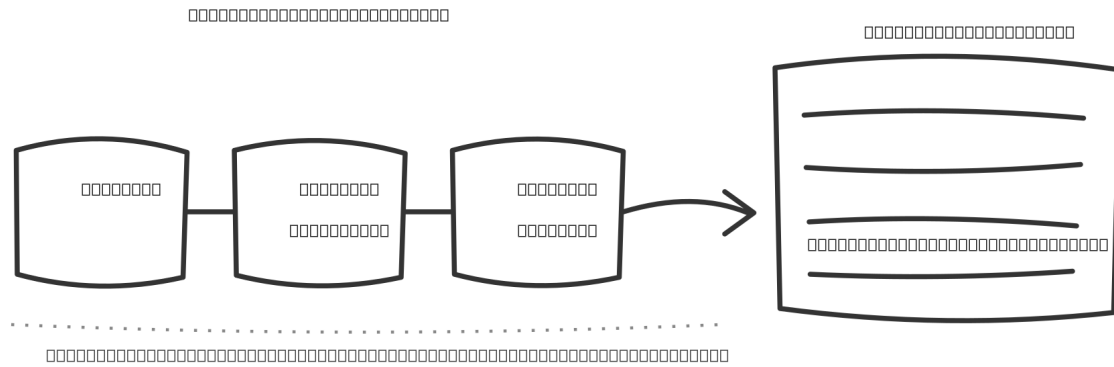
3 Keep time supplied and ordering explicit

The policy reads no clock. Each call supplies `now`; every record retains its observation time and nonzero sequence. Age and gap at their exact configured maxima pass. One tick beyond fails. Future, backward, and exact half-range time are explicit failures. Sequence rollover from `UINT32_MAX` to 1 is valid because zero is reserved. A fieldwise-identical duplicate is idempotent and does not increase the count. A changed duplicate, gap, regression, zero, or half-range ambiguity ends the attempt with attributable evidence.

4 Let every record prove stationary behavior

Mapping happens before bounds checks. Each mapped acceleration component must remain inside its inclusive deviation around the configured expected stationary acceleration. Each mapped angular-rate component must remain between its inclusive positive and negative limit.

Every individual record must pass. A plausible average cannot conceal a single moving, saturated, stale, or faulted record. Exact limits pass; one unit beyond fails. Differences, absolute values, sums, and means use checked widened arithmetic.



Alternative text: record cards advance through a bounded collecting window to a terminal card; the first qualification or precise rejection freezes the attempt until reset.

Point	Expected state	Count effect
begin	Collecting	zero
$N - 1$ accepted	Collecting	advances once per new record
N th accepted	Qualified	terminal at N
first bad record	Rejected	no convenient replacement
later record	unchanged terminal result	none

5 Separate operation failure from unhealthy evidence

Malformed configuration, invalid record structure, or lifecycle misuse returns a non-OK **Status** and changes nothing. A well-formed but unhealthy record is meaningful domain evidence: it may return operation status OK while publishing **Rejected** and a precise qualification reason.

6 Apply one deterministic precedence

The implementation stages the complete candidate before changing accepted count, extrema, sums, or evidence. When failures collide, the same ordered rule always chooses the scalar reason while the terminal record retains the complete collision.



Alternative text: a graphite checklist orders lifecycle, source domain, chronology, producer quality, mapping, numeric bounds, and accumulation from validate-first to commit-once.

A producer fault dominates not-ready and saturation because its numeric payload is not trusted. With producer status OK, not-ready dominates saturation; saturation dominates stationary arithmetic. Acceleration out-of-window supplies the reason before angular rate when both fail.

Atomicity experiment. Place recognizable canaries around the evidence object. Submit a changed duplicate, an unrepresentable mapping, and a post-terminal record. Verify that each rejected operation leaves the previous evidence and both canaries byte-for-byte unchanged.

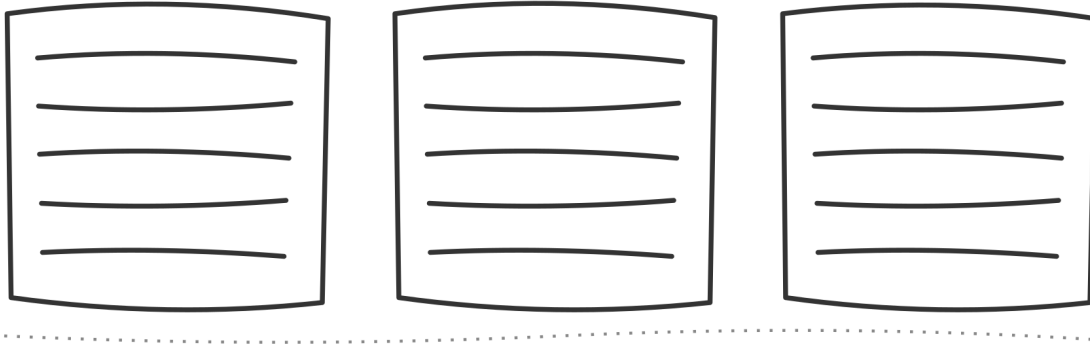
7 Read the complete evidence, not a Boolean

The copied envelope retains attempt and lifecycle generation, qualification revision, mapping, state and reason, accepted count, first/last sequences and times, maximum age and gap, mapped means, extrema and widened sums, the terminal source-frame record, mapped terminal record, and status.

Lesson 069 consumes this complete immutable envelope. It must not combine a record from one attempt with qualification from another or remap a mutable child snapshot.

8 Replay two traces without touching hardware

The Mega sketch follows acquire, configure, start, then observe, decide, actuate. It normalizes copied synthetic samples, qualifies one three-record stationary trace, resets explicitly, and rejects one independently faulted trace. “Actuate” means publish inert memory values.



Alternative text: three sketched cards prompt the learner to predict state and count, observe named volatile evidence cells, and interpret one qualifying trace plus one rejected fault trace without a hardware claim.

1. **Predict** the state, reason, count, source ID, and first/last sequence before each call.
2. **Observe** the two named volatile result cells and completion cell with a debugger, simulator, or disassembly. Serial is optional.
3. **Interpret** the stationary result as E0 contract evidence and the producer-fault result as proof that unhealthy input was retained.

The memory cells are the non-Serial observation path. Resource-acquisition evidence is separately zero pins, timers, interrupts, buses, endpoints, sensors, and supplies. Safe-state evidence is the absence of hardware intent. Neither proves that a physical unit is attached, stationary, or safe to power.

9 Test the finite matrix

- Classify all 216 mappings with an independent determinant oracle: exactly 24 pass.
- Exercise every identity/revision mismatch, producer state, readiness, saturation, signed extreme, and mapping failure.
- Test window sizes 2 and 32; exact and one-over age, gap, acceleration, and rate limits; duplicates, rollover, regression, and collisions.
- Test reset and shutdown at every count, terminal immutability, output canaries, and replay equivalence after Lesson 067 decode.

10 Troubleshooting matrix

Symptom	Inspect first	Interpretation
ConfigurationMismatch	complete source and revision domain	a familiar model is not enough
ProducerFault	terminal record status	numeric payload was not trusted
NotReady	status, ready flag	valid report, not an accepted sample
Saturated	declared range and saturation	excursion cannot enter averages
Stale	supplied now and observed time	maximum age is inclusive
sequence discontinuity	previous/current sequence	duplicate, gap, regression, or ambiguity
timestamp discontinuity	previous/current observed time	maximum gap is inclusive
numeric rejection	mapped vectors and per-axis limits	every axis of every record must pass

11 Exercises

1. Explain why source selection means configuration, not runtime probing.
2. Design a collision where producer fault, not-ready, and saturation are all present. Which reason wins, and what evidence remains visible?
3. Explain why a passing mean cannot excuse one acceleration excursion.
4. Describe an identical duplicate and a changed duplicate. Why does only the first leave the attempt collecting?
5. List the exact E1 evidence needed before claiming support for one physical MPU6050 or QMI8658 carrier.

12 Open acceptance record

Host evidence	Record strict tests, sanitizers, all 216 mappings, lifecycle, precedence, boundaries, canaries, and decoded replay equivalence.
Mega evidence	Compile the canonical synthetic replay; record exact flash, static SRAM, and synchronous stack evidence.
Resource evidence	Target policy plus result at most 384 bytes and stack at most 384 bytes; 512 bytes is the independently reviewed ceiling for either.
Document evidence	Build twice; inspect metadata, embedded fonts, extraction, monochrome output, links, visual classifications, and pencil legibility.
Physical evidence	Open E1: exact specimen, electrical topology, register behavior, mounting, accuracy, stationary fixture, and bench acceptance.

Acceptance item	Record
Commit and toolchain	_____
Host / sanitizer / mapping result	_____
Flash / SRAM / stack / object sizes	_____
Replay and non-mutation result	_____
PDF visual and extraction review	_____
Open E1 owner and next action	_____

Stop condition. Do not call this sensor qualification. A completed E0 exercise proves only that one configured stream of copied records met a deterministic stationary contract. Physical identity, acquisition, mounting, calibration, observation, and safe state remain separate acceptance work.

Next: Lesson 069 — Record one qualified source session.